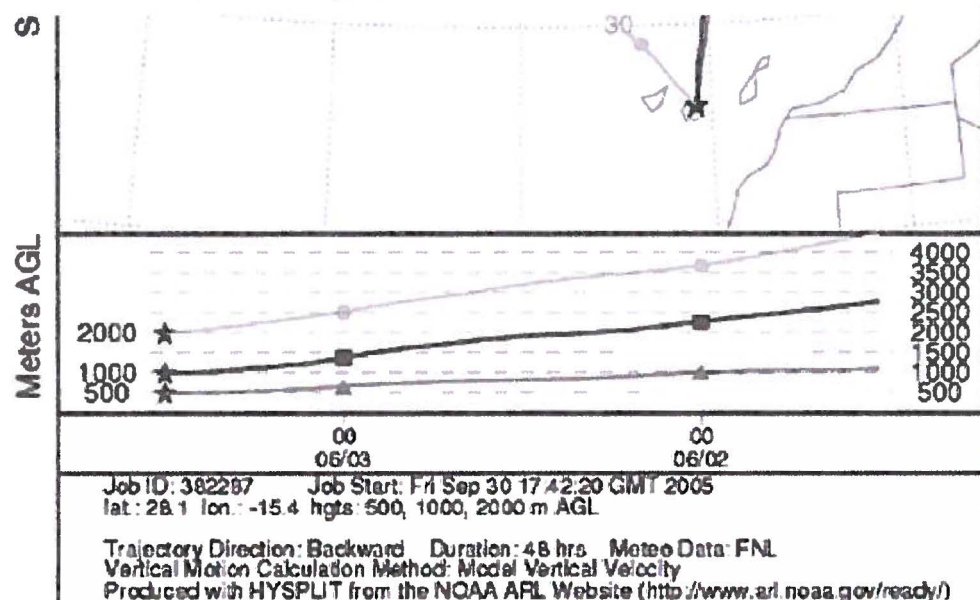




The number of species found in SD #15 was lower than that of SD #7 (5 and 13, respectively) and showed two species with a higher number of clones per species: *Staphylococcus kloosii/carnosus* (6) and *Bradyrhizobium* sp. (4) (Table 1.0). We suggest that the high number of clones per species, identified in SD#15, represent lower diversity. In contrast, in SD#7, we identified a lower number of clones per species, but the representative genera was richer. In conclusion, the number of clones per species was inversely proportional to the identified bacterial diversity. Fifty percent (50%) of the bacteria identified in SD#7 are associated with human health issues. These include: *Pseudomonas*, *Rahnella aquatilis*, *Serratia fonticola*, *Sphingomonas*, *Shigella flexneri*, and *Corynebacterium mucifaciens*. In contrast, only one of the species, *Corynebacterium mucifaciens*, identified in SD #15 is pathogenic to humans. The other four species identified from SD#15 are part of normal or resident microbiota of animals (*Staphylococcus kloosii*; *Propionibacterium acnes*), soil (uncultured soil bacterium) or plants (*Bradyrhizobium*). Interestingly, by comparing the identified species, it is clear that all the species identified in the light dust event sample (SD#15) were also present in the heavy dust event sample (SD#7).

Sahara Dust Sample #7			
No. of Clones	Matching organism(s)	GenBank Accession number	% homology
1	Arctic sea ice bacterium ARK9973/Marine sediment bacterium OC-12 <i>Bradyrhizobium</i> sp.	DQ453581	99 a
2	<i>Corynebacterium mucifaciens</i>	DQ453573, DQ453583	99
1	<i>Modestobacter</i> sp. Elin 165 a	DQ453589	99
1	<i>Morganella morganii</i>	DQ453582	97 a
2	<i>Propionibacterium acnes</i>	DQ453569, DQ453572	100
2	<i>Pseudomonas</i> sp.a	DQ453576, DQ453579	98
1	<i>Rahnella aquatilis</i> isolate k	DQ453577	99 a
1	<i>Serratia fonticola</i>	DQ453570	100
1	<i>Shigella</i> sp/ <i>Escherichia</i> sp	DQ453578	99
1	<i>Sphingomonas</i> sp.	DQ453585	100
4	<i>Staphylococcus kloosii/carnosus</i>	DQ453571, DQ453584, DQ453586, DQ453588	97-99
3	Uncultured bacterium clone MP104-01	DQ453574, DQ453575, DQ453580	99
1	<i>Sphingomonas</i> sp.	DQ453587	93 a
Sahara Dust Sample #15			
No. of Clones	Matching organism(s)	GenBank Accession number	% homology
4	<i>Bradyrhizobium</i> sp.	DQ453590, DQ453598, DQ453601, DQ453593	99
1	<i>Corynebacterium mucifaciens</i>	DQ453596	99
1	<i>Propionibacterium acnes</i>	DQ453597	98
6	<i>Staphylococcus kloosii/carnosus</i>	DQ453594, DQ453599, DQ453600, DQ453592, DQ453602, DQ453603	99
1	Uncultured bacterium clone mw5	DQ453595	99
8	No Insert	False Positives	